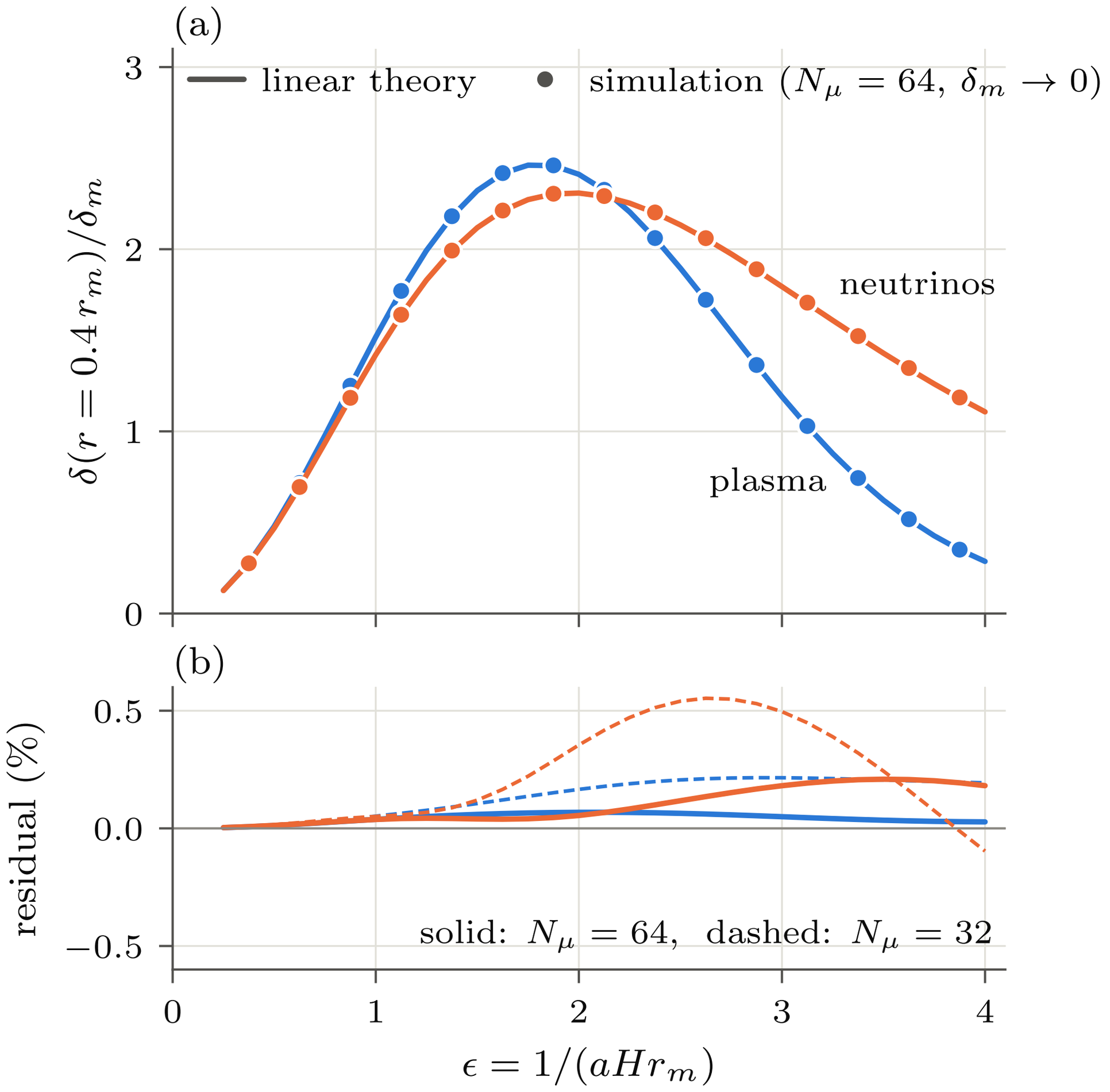

(a)
linear theory
simulation ($N_\mu = 64$, $\delta_m \to 0$)
neutrinos
plasma
$\delta(r = 0.4\,r_m)/\delta_m$
3
2
1
0
(b)
residual (%)
0.5
0.0
−0.5
solid: $N_\mu = 64$, dashed: $N_\mu = 32$
0
1
2
3
4
$\epsilon = 1/(aHr_m)$